Beyond Unitary Designs: Quantifying Trainability in Parameterized Quantum Circuits

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(Dated: March 31, 2026)

Parameterized quantum circuits (PQCs) represent a central class of controllable quantum evolutions, yet their trainability can be severely limited by exponentially vanishing gradients, known as the barren plateau phenomenon. Existing analyses typically rely on strong randomness assumptions, such as unitary designs and remain limited in addressing general structured circuits. In this Letter, we develop a measure-based framework to characterize the trainability of general PQCs and establish a level classification of gradient variance across orthogonal subspaces. Our results reveal a trainability advantage in previously unexplored unitary 1-designs compared to higher-order designs. We further introduce the improvement potential to quantify maximal variance enhancement under local disturbances. This framework provides unified analytical guidance for understanding and engineering trainable quantum circuits on near-term quantum hardware.

After decades of development, quantum technology has demonstrated explicit advantages in areas such as accelerated computing and precision measurement [1–5]. As a trainable framework for realizing such advantages, PQCs have become a widely used architecture for qubit computation and evolution, with notable applications in quantum metrology [6–9] and variational quantum algorithms (VQAs) [10–13]. PQC's ability to solve given problems stems from its tunable internal parameters, which allow us to optimize them within a fixed ansatz to accomplish tasks. In practical applications, such optimization is typically achieved using gradient descent and its variants. However, it has been shown that for a broad class of parametrized quantum circuits, particularly those that are universal, the gradient of the cost function will vanish exponentially with the size of the system[14, 15], resulting in severe trainability issues and degraded optimization performance, known as the barren plateau. Therefore, for a given PQC architecture, evaluating the probability of its gradients vanishing exponentially under the current parameter distribution is a crucial topic in the study of circuit trainability.

Gradient variance is widely used as a standard indicator of gradient vanishing. By Chebyshev's inequality, a small gradient variance implies a high probability of vanishing gradients[16], which reduces the trainability problem to the evaluation of gradient variances under a specified parameter distribution. Most existing studies evaluate gradient variances under the unitary 2-design assumption[17–20], where a unitary t-design denotes an ensemble of unitary operators whose statistical moments up to order t coincide with those of the Haar random measure. The Haar measure enables a precise evaluation of gradients for circuits that approximate a global unitary 2-design. However, this assumption is only satisfied by PQCs with strong randomness. Such circuits exhibit gradient variances scaling as d^{-2}(with Hilbert space dimension d), leading to exponentially flat optimization landscapes and poor convergence. For circuits that do not satisfy this assumption, a reliable analytical framework remains lacking. Recent research has extended this analytical framework to cover circuits composed of multiple local unitary 2-designs; nevertheless, each local 2-design remains constrained by the strong randomness assumption[21, 22].

In practice, quantum circuits are often designed with tailored structures for specific problems, yet they typically lack sufficient randomness and thus fall outside the scope of existing analytical frameworks. Since such non-random PQCs play a central role in practical quantum circuit design and applications, it is essential to develop a general, computable gradient variance framework that can adapt to all types of PQCs.

In this Letter, we develop a measure-based framework to characterize the trainability of PQCs beyond the standard unitary design paradigm. By introducing a class of polynomial probability measures, we prove that the distribution of any PQC can be approximated with arbitrary precision while preserving its gradient variance. Using zonal polynomials and Weingarten calculus [23–25], we obtain explicit variance expressions and establish a level classification of trainability across orthogonal subspaces (Theorem 2). In particular, this framework identifies a significant trainability advantage in previously overlooked unitary 1-designs compared to higher-order designs. Furthermore, we define the concept of improvement potential to quantify the maximal variance enhancement achievable under local disturbance (Theorem 3), thereby providing a criterion for designing more trainable PQCs. These results provide analytical guidance to identify and enhance the trainability of parameterized quantum circuits in the noisy intermediate-scale quantum(NISQ) era.

Preliminaries.— The performance of VQAs is related to the selected ansatz in the training. Generally, an ansatz consists of a PQC $U(\boldsymbol{\theta})$, and a cost function $C(\boldsymbol{\theta})$ with trainable parameters $\boldsymbol{\theta}$. Since trainability depends on the type of selected ansatz, we adopt the hardware-efficient ansatz as our foundational framework, which has been widely studied in theoretical research on barren

plateaus [12, 26–28]. In a hardware-efficient ansatz, a PQC with L layers is given by $U(\boldsymbol{\theta}) := U(\theta_1, \cdots, \theta_L) = \prod_{l=1}^{L} U_l(\theta_l) W_l$, where $U_l(\theta_l) = \exp(-iH_l\theta_l)$ is a rotation gate and W_l is the entangling gate chosen from a fixed set of unitary operators. The cost function is of the form $C(\boldsymbol{\theta}) = \mathrm{Tr}[U(\boldsymbol{\theta})\rho U^\dagger(\boldsymbol{\theta})O]$. Here, ρ is the input density matrix, O is the target observable, and the rotation angles $\{\theta_j\}$ are typically assumed to be independent. Following the standard research on barren plateaus [14, 16], we characterize trainability by the variance of the gradient $\mathrm{Var}_{\boldsymbol{\theta}}[\partial_\ell C] := \mathrm{Var}_{\boldsymbol{\theta}}[\partial C/\partial\theta_\ell]$.

To analyze the structure of the partial derivative $\partial_\ell C$, we split the circuit at the parameter θ_ℓ into two parts, U_L and U_R, i.e., $U = U_L U_R$, where $U_L(\boldsymbol{\theta}) = \prod_{j=\ell+1}^{L} U_j(\theta_j) W_j$, $U_R(\boldsymbol{\theta}) = \prod_{i=1}^{\ell} U_i(\theta_i) W_i$. Consequently, the partial derivative $\partial_\ell C$ can be expressed as

$$\partial_\ell C = i\,\mathrm{Tr}[U_R \rho U_R^\dagger [H_\ell, U_L^\dagger O U_L]]. \tag{1}$$

Let $X_L = i[H_\ell, U_L^\dagger O U_L]$, then the gradient variance can be expressed explicitly as

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_\ell C] = \int \mathrm{d}\mu_L(U_L)\,\mathrm{d}\mu_R(U)\,\mathrm{Tr}[U\rho U^\dagger X_L]^2. \tag{2}$$

Since the form of $\partial_\ell C$ is associated with the 1-moment of U_R and U_L, the concept of unitary designs is naturally applied to the evaluation of gradient variances. Formally, a probability measure μ is said to form a unitary t-design[29, 30] if, for every homogeneous polynomial $p \in \mathrm{Hom}(U(d), t, t)$, we have $\int \mathrm{d}\mu(U)p(U) = \int \mathrm{d}\mu_H(U)p(U)$, where $\mathrm{Hom}(U(d), t, t)$ denotes the set of all polynomials that are homogeneous of degree t in both the matrix entries of U and $U^\dagger$, and μ_H is the normalized Haar measure.

Polynomial approximation to general PQC.—As previously discussed, the local unitary 2-design assumption is not fit for many practical PQCs. A new framework is required to develop a general theory of trainability. In the following, we introduce the framework of balanced polynomial measures, which serves as the foundation for our work.

When a PQC is equipped with a specific parameter distribution, it induces a probability measure $\mu \in \mathcal{P}(U(d))$. We call a finite measure μ a polynomial measure if the Radon–Nikodym derivative [31] $\mathrm{d}\mu/\,\mathrm{d}\mu_H$ is a polynomial function. Let $\mathcal{G}(U(d))$ denote the set of all real polynomial measures. Since the set $\mathcal{I}(U(d)) := \mathcal{P}(U(d)) \cap \mathcal{G}(U(d))$ is weak*-dense in $\mathcal{P}(U(d))$, it leads to the universal approximation of arbitrary PQCs by polynomial probability measures. Let $\mathrm{Var}_{\boldsymbol{\theta}}^{(\mu_L,\mu_R)}[\partial_\ell C(\theta)]$ denote the gradient variance with $U_L \sim \mu_L$ and $U_R \sim \mu_R$, i.e., each drawn from the corresponding probability measure. Then, two close probability measure pairs (μ_L, μ_R) and (μ_L', μ_R') would lead that $\mathrm{Var}_{\boldsymbol{\theta}}^{(\mu_L,\mu_R)}[\partial_\ell C]$ are close to $\mathrm{Var}_{\boldsymbol{\theta}}^{(\mu_L',\mu_R')}[\partial_\ell C]$. This compatibility implies that the trainability of arbitrary PQCs can be characterized by integrals of polynomial functions with respect to the Haar measure, whose evaluation is enabled by Weingarten calculus. Figure 1 illustrates the relative positions of the covered spaces of unitary 2-designs and polynomial measures in $\mathcal{P}(U(d))$.

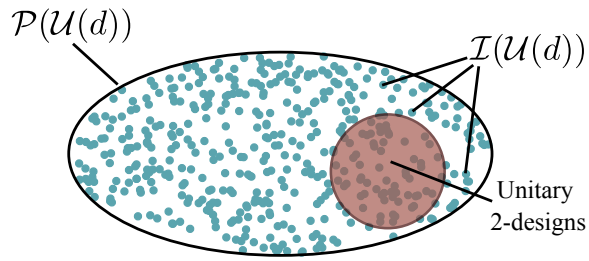

FIG. 1: Schematic illustration of the relative positions of polynomial probability measures $\mathcal{I}(U(d))$ and unitary 2-designs within the space of probability measures $\mathcal{P}(U(d))$. The green discrete points represent elements of $\mathcal{I}(U(d))$, which are densely distributed throughout $\mathcal{P}(U(d))$, while the red region denotes the set of unitary 2-designs, concentrated in a small subset of $\mathcal{P}(U(d))$.

In the polynomial measure $\mathrm{d}\mu_R/\mathrm{d}\mu_H$, only a portion of its components contribute to the gradient variance. Every monomial in these components on $\mathcal{U}(d)$ belongs to the space $\mathrm{Hom}(\mathcal{U}(d), t, t)$ for some $t \in \mathbb{Z}_+$, and we refer to them as balanced components. Balanced components can be expanded as linear combinations of zonal polynomials $Z_{\boldsymbol{\lambda}}(U)$, labeled by indices $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \ldots, \lambda_d)$ satisfying $\lambda_i \in \mathbb{Z}$ and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$. When the balanced components in the measure are expanded via zonal polynomials, the total gradient variance is consequently decomposed into a sum of several components. Notably, the orthogonality properties of zonal polynomials restrict the contributing terms to those with indices lying in the set $S = \{\boldsymbol{\lambda} : |\boldsymbol{\lambda}| = 0, |\boldsymbol{\lambda}_+| \leq 2\}$, where $|\boldsymbol{\lambda}_+|$ denotes the sum of the positive entries of $\boldsymbol{\lambda}$.There are $|S| = 6$ such types of zonal polynomials, whose explicit forms are detailed in [32].

If we further assume that each parameter satisfies $\theta_i \sim \mathrm{Uniform}[0, 2\pi)$, then the polynomial measures to approximate each component of PQC (including single rotation gates and fixed entangling gates) can be exactly obtained by the following theorem:

Theorem 1. *Let $R = \exp(iP\theta)$ be a rotation gate, where $P = \otimes_{i=1}^{n}\sigma_i$ is a Pauli word and $\theta \sim \mathrm{Uniform}[0, 2\pi)$. The distribution of R admits an approximation by the following partial sums*

$$p_N = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\theta \sum_{|\boldsymbol{\lambda}|\leq N} Z_{\boldsymbol{\lambda}, e^{iP\theta}}, \quad N \in \mathbb{Z}_+. \tag{3}$$

Furthermore, at the level of computing gradient variance,

the distribution of R is equivalent to

$$p_{2,b} = 1 + \frac{1}{2}(Z_{(1,0,\cdots,0,-1),I} + Z_{(1,0,\cdots,0,-1),P}) + \frac{1}{4} \sum_{|\boldsymbol{\lambda}|=0,\ |\boldsymbol{\lambda}_+|=2} (Z_{\boldsymbol{\lambda},I} + Z_{\boldsymbol{\lambda},P} + Z_{\boldsymbol{\lambda},\frac{(I+iP)}{\sqrt{2}}} + Z_{\boldsymbol{\lambda},\frac{(I-iP)}{\sqrt{2}}}). \tag{4}$$

For a fixed entangling gate W_ℓ, its distribution is given by the Dirac delta distribution concentrated at W_ℓ.

The probability measure of the whole circuit is just the convolution of the probability measures of each component in order. Thus, we can summarize the following corollary:

Corollary 1.1. *Let $U(\boldsymbol{\theta}) = \prod_{j=1}^{L} \exp(iP_j\theta_j)W_j$, where P_j are Pauli words and $U \sim$ Uniform$[0, 2\pi)$ independently. Then, the gradient variance induced by the circuit U coincides with that of the polynomial f defined below.*

$$f = 1 + \frac{1}{2^L} \sum_{i_1,\ldots,i_L=1}^{2} Z_{(1,0,\ldots,0,-1),\prod_{s=1}^{L} P_s^{i_s} W_s} + \frac{1}{4^L} \sum_{|\boldsymbol{\lambda}|=0,\ |\boldsymbol{\lambda}_+|=2} \sum_{i_1,\ldots,i_L=1}^{4} Z_{\boldsymbol{\lambda},\prod_{s=1}^{L} P_s^{i_s} W_s}. \tag{5}$$

Here, for a Pauli word P, we write $P^1 := I,\ P^2 := P,\ P^3 := (I + iP)/\sqrt{2},\ P^4 := (I - iP)/\sqrt{2}$.

Corollary 1.1 provides an explicit polynomial measure family to approximate any PQC under the hardware-efficient ansatz. Thus, the gradient variance of an arbitrary PQC can be obtained by the Weingarten calculus of polynomials with respect to the Haar measure. For rigorous approximation statements and proofs of Theorem 1 and Corollary 1.1, one can refer to the Supplementary Materials Sec. II.

In the remaining content, we will delve deeper into the analysis of variance on the balanced polynomial measures. Since the analysis for U_L and U_R is analogous, without loss of generality, we focus on the right part U_R in the discussion[33].

Level classification of variances.— The above analysis shows that an arbitrary PQC can be approximated by polynomial probability measures. The balanced component of such measures, which governs the gradient variance, admits a decomposition in terms of zonal polynomials [34–36]. Consequently, the total variance can be expressed as the sum of three contributions:

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) = \sum_{i=0}^{2} \mathrm{Var}_i^{(\mu_L,\mu_R)}(\rho, O). \tag{6}$$

Through the zonal polynomial decomposition, we have projected the variance onto several subspaces that admit explicit computation successfully. Among these, the bound $\mathrm{Var}_0 \le C_0(\rho, O)d^{-2}, C_0(\rho, O) > 0$ for the zeroth-order variance is a well-established result. For the first- and second-order variances, we provide the following theoretical upper bounds.

Theorem 2. *Given a balanced probability polynomial measure μ_R, the α-th-order variance admits the following asymptotic bounds:*

$$\mathrm{Var}_\alpha^{(\mu_L,\mu_R)}(\rho, O) \le C_\alpha(\rho, O)d^{\alpha-2} \quad \text{for } \alpha = 0, 1, 2, \tag{7}$$

where $C_\alpha(\rho, O)$ is a constant independent on d.

Building on the uniform approximation by polynomial measures, Theorem 2 characterizes the variance contributions from each orthogonal subspace for general PQCs, which provides a level classification of total variances based on structural properties of measures. It demonstrates that the second-order variance determines whether there exists a barren plateau at the given parameter. Moreover, for a pure state ρ, there is a further restriction that

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) \le \frac{d}{2}\mathrm{Var}_1^{(\mu_L,\mu_R)}(\rho, O) + \mathcal{O}(d^{-1}), \tag{8}$$

indicating that a large first-order variance is necessary to avoid degraded trainability.

From Eq. (8), we can derive that when at least one of μ_L or μ_R forms a unitary 1-design, the total variance is upper-bounded by d^{-1} up to a constant. The following proposition shows the attainability of this bound.

Proposition 1. *Let ν_m be a probability measure on the unitary group $U(d)$ whose derivative satisfies*

$$\frac{\mathrm{d}\nu_m}{\mathrm{d}\mu_H}(U) = c_m \,\mathrm{Tr}[U\rho U^\dagger P]^{2m}, \tag{9}$$

where ρ is a pure state, P is an idempotent matrix, and c_m is the normalization constant. Then ν_m defines a unitary 1-design. Moreover, for sufficiently large m, there exists a measure μ_L and an observable O such that

$$\mathrm{Var}^{(\mu_L,\nu_m)}(\rho, O) \ge \gamma d^{-1} \tag{10}$$

for some constant $\gamma > 0$.

This result indicates that when the PQC forms a unitary 1-design, its total gradient variance can be at most on the order of d^{-1}, thereby exhibiting superior trainability compared to unitary 2-designs, which are bounded by the d^{-2} scaling. The proofs of Theorem 2 and Proposition 1 are provided in Supplementary Material, Sec. III.

Furthermore, we demonstrate the attainability of the d^{-1} scaling of the gradient variance at a given parameter θ_ℓ under a specific unitary 1-design setting. Consider the rotation gate at position θ_ℓ to be $\exp(iP_1\theta_\ell)$ and the observable $O = P_2/\sqrt{d}$, where P_1 and P_2 are two non-commuting Pauli words and U_L is an operator commuting

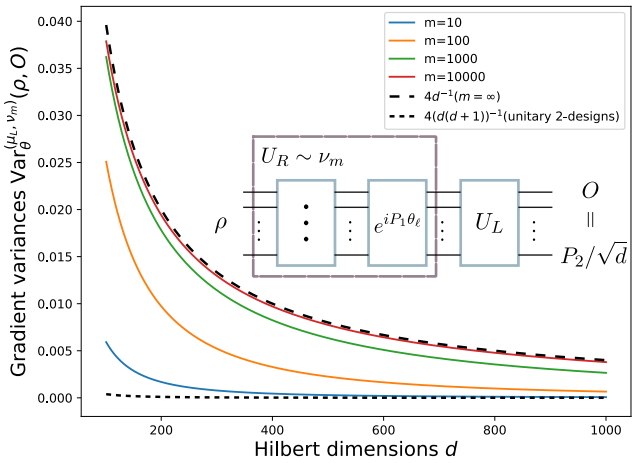

FIG. 2: The gradient variances of the circuit when adopting unitary 1-designs and unitary 2-designs as functions of the system size d. The inset shows the circuit structure used in this example. The long-dashed line represents the upper bound $4d^{-1}$ for unitary 1-designs in this circuit and the colored solid lines correspond to unitary 1-designs with μ_R taking different ν_m. The gradient variance gradually approaches the upper bound as m increases. In contrast, when the circuit adopts a unitary 2-design, the gradient variance is fixed at $4(d(d+1))^{-1}$ (short-dashed line).

with P_2. When U_R forms a unitary 2-design, the variance $\mathrm{Var}^{(\mu_L,\mu_R)}(\rho,O)$ can be computed as $4(d(d+1))^{-1}$, as indicated by the short-dashed line in Figure 2. When U_R is taken as the unitary 1-design satisfying Eq. (9) with $[P_1,P_2]=2iP$, the variance $\mathrm{Var}^{(\mu_L,\mu_R)}(\rho,O)$ is bounded by the d^{-1} scaling and, as m increases (colored solid lines in Figure 2), approaches the upper bound $4d^{-1}$, marked by the long-dashed line.

The Improvement Potential.— So far, several methods have been developed to enhance trainability, such as the substitution by shallow circuits, tailored problem-specific ansatz, and optimized initialization strategies [17, 37–39]. However, these approaches typically rely on global or heuristic modifications and provide limited quantitative guidance within a fixed circuit structure. Here, we propose a formula to characterize the maximal improvement of trainability through inserting oracles adjacent to a given position. Specifically, we insert a parametric oracle S_p associated with the probability measure μ on the left side of U_R. The probability measure for the combination of U_R and S_p is the convolution $\mu * \mu_R$. The schematic diagram of the process is shown in Figure 3.

Formally, we consider the following quantity

$$k_{\mu_R} := \sup_{\mu\in\mathcal{P}(U(d))} \frac{\mathrm{Var}^{(\mu_L,\mu*\mu_R)}(\rho,O)}{\mathrm{Var}^{(\mu_L,\mu_R)}(\rho,O)}, \tag{11}$$

which characterizes the maximal relative enhancement of variance at the parameter θ_ℓ by inserting an oracle.

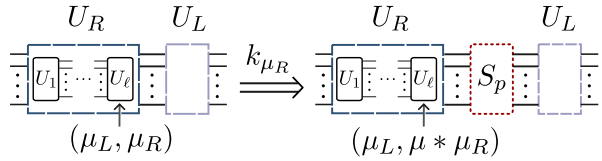

FIG. 3: The process of inserting oracles adjacent to the given position to potentially enhance trainability. Here, S_p denotes an arbitrary unitary oracle associated with probability measure μ and the improvement potential k_{μ_R} represents the maximal improvement of the total variance during the process.

If $k_{\mu_R}=1$, the PQC exhibits no improvement potential at θ_ℓ; whereas $k_{\mu_R}>1$ indicates the existence of improvement potential at that parameter. We provide the following criterion to determine whether $k_{\mu_R}>1$ or not:

Theorem 3. *For a PQC at θ_ℓ with the probability measure pair (μ_L,μ_R), the improvement potential obtained by inserting an oracle is*

$$k_{\mu_R} = \frac{\|g*h\|_\infty}{\|gh\|_1}. \tag{12}$$

Here g and h are functions on $U(d)$ defined by

$$g(V) = \frac{\mathrm{d}\mu_R}{\mathrm{d}\mu_H}(V), h(V) = \int \mathrm{d}\mu_L(U_L)\ \mathrm{Tr}\left[V^\dagger\rho V X_L\right]^2. \tag{13}$$

The function g denotes the derivative of μ_R with respect to μ_H, while h represents the variance obtained after integrating over the left unitary. Both functions can be decided by the pair (μ_L,μ_R).

Moreover, we find that the quantity $\mathrm{Var}^{(\mu_L,\mu_H)}(\rho,O)$ can serve as a reference threshold for determining whether improvement potential exists at a given parameter, where this value corresponds to the unitary 2-design case. For a pair of probability measures (μ_L,μ_R), if $\mathrm{Var}^{(\mu_L,\mu_R)}(\rho,O) < \mathrm{Var}^{(\mu_L,\mu_H)}(\rho,O)$, then the pair possesses improvement potential. In contrast, for circuits where μ_R satisfies the unitary 2-design property, the gradient variance satisfies $\mathrm{Var}^{(\mu_L,\mu_R)}(\rho,O) = \mathrm{Var}^{(\mu_L,\mu_H)}(\rho,O)$. However, such circuits exhibit no training potential, i.e., $k_{\mu_R}=1$. Therefore, all parameterized quantum circuits, after being optimized through the process illustrated in Figure 3, can achieve performance at a given parameter position that is no worse than that of unitary 2-design circuits. When $\mathrm{Var}^{(\mu_L,\mu_R)}(\rho,O) \geq \mathrm{Var}^{(\mu_L,\mu_H)}(\rho,O)$, no definitive conclusion can be drawn from this comparison alone. In this regime, the presence or absence of improvement potential must be determined using the criterion established in Theorem 3.More detailed discussions and constructive examples are provided in the Supplementary Material Sec. IV.

Numerical Simulation.—To validate our theory, we perform numerical simulations using Monte Carlo methods for circuits satisfying the unitary 1-design condition, computing the total gradient variance at the parameter θ_ℓ, and compare the results with the theoretical predictions given by our polynomial method. The simulated circuit structure is illustrated in Figure 4a, where U_R is a universal block constructed following the approach in [40, 41], with its measure derivative satisfying Eq. (9). The left unitary $U_L = \exp(iP'\beta)$ is an arbitrary rotation gate, where P' denotes a Pauli word.

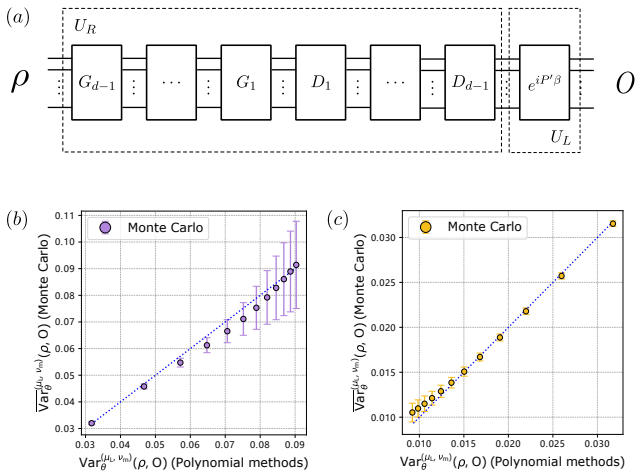

FIG. 4: Simulation setup and validation results. (a) The quantum circuit used in simulations. Here, G_k and D_k represent rotation gates of different structures and the gradient variance is evaluated at D_{d-1}. See Supplemental Material Sec. V for detailed parameter configurations. (b), (c) The correlation of theoretical predictions of gradient variance obtained from our method and Monte Carlo estimates on a three-qubit system with the observable $O = Z \otimes I \otimes Z$ and the initial pure state $\rho = |0\rangle\langle 0|$. Each point corresponds to a different value of the polynomial degree $2m$ and the dashed line is $y = x$, which indicates perfect agreement. In (b), $P = X \otimes Z \otimes Z$. In (c), $P = Z \otimes Z \otimes Z$.

Although our theory applies to circuits of arbitrary size, numerical verification is restricted to the three-qubit case due to the computational complexity of Monte Carlo methods. Figures 4b and 4c show correlation plots of a series of the theoretical gradient variances $\mathrm{Var}^{(\mu_L,\nu_m)}(\rho, O)$ and the Monte Carlo estimates $\overline{\mathrm{Var}}^{(\mu_L,\nu_m)}(\rho, O)$, both obtained by varying the balanced polynomial degree $2m$ of ν_m for two different choices of U_L. The theoretical values are computed using our polynomial method, while the estimates are obtained via Monte Carlo simulation. In both cases, the data points cluster tightly around the line $y = x$, and the differences between the theoretical values and the estimates are within one standard deviation, showing excellent agreement.

Trainability plays a central role in the optimization of PQCs, as it determines whether circuit parameters can be efficiently updated. In this Letter, we introduce a measure-theoretic framework based on balanced polynomial measures to characterize the trainability of PQCs. We show that balanced polynomial measures can approximate any PQC with arbitrary precision up to a global factor. Within this framework, we propose a level classification of trainability based on zonal polynomial decomposition. As an application, we rigorously demonstrate that unitary 1-designs can exhibit superior trainability compared to unitary 2-designs. These theoretical findings are further supported by numerical simulations. Furthermore, we introduce the concept of improvement potential and identify a reference threshold for determining whether the trainability of a circuit possesses such potential, and present illustrative examples for different scenarios. Our work reveal internal factors that govern trainability and provide guiding principles for designing more trainable PQCs. Future research may benefit from the decomposition of the space that circuits occupy using dynamic Lie algebras [42, 43], which could provide a more practical and structurally informed framework for achieving more precise approximations.

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